

than” the other one. This could mean that one file’s assigned number is higher than the other’s assigned number, or it could mean that it’s higher in an alphabetical or other ordering.

1. Select the highest file in the filing cabinet. (We’ll start at the back of the cabinet and work our way to the front.)
2. Compare the file you have selected with the file to insert.
3. If the file you have selected is lower than the file to insert, place the file to insert one position behind that file.
4. If the file you have selected is higher than the file to insert, select the next highest file in the filing cabinet.
5. Repeat steps 2 to 4 until you have inserted your file or compared it with every existing file. If you have not yet inserted your file after comparing it with every existing file, insert it at the beginning of the filing cabinet.

That method should more or less match the intuition you have for how to insert a record into a sorted list. If you prefer, you could also start at the beginning of the list, instead of the end, and follow an analogous process with the same results. Notice that we haven’t just inserted a record; we’ve inserted a record *in the correct position*, so after insertion, we’ll still have a sorted list. We can write a script in Python that executes this insertion algorithm. First, we can define our sorted filing cabinet. In this case, our filing cabinet will be a Python list, and our files will simply be numbers.

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```
cabinet = [1,2,3,3,4,6,8,12]
```

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Then, we can define the “file” (in this case, just a number) that we want to insert into our cabinet.

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```
to_insert = 5
```

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We proceed one at a time through every number in the list (every file in the cabinet). We’ll define a variable called `check_location`. As advertised, it will store the location in the cabinet that we want to check. We start at the back of the cabinet:

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```
check_location = len(cabinet) - 1
```

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